Pure discrete spectrum for Pisot substitution tilings

Version 15

Abstract

We study the Pisot substitution conjecture in the standard formulation: every primitive Pisot substitution whose incidence matrix has irreducible characteristic polynomial gives rise to a tiling dynamical system with pure discrete spectrum. Our contribution is a structural simplification of Levels 2 and 3 of the balanced-pair route. The engine is unique decodability of the σ-image code, a five-line corollary of the classical defect theorem combined with $\det M_\sigma \neq 0$. In Level 2 it gives a finite phase-automaton diagnostic for coincidence padding: complete-cutting non-diagonal cycles are ruled out by unique decodability, while nonzero-offset recurrences are not controlled; finiteness of the balanced-pair automaton is carried as a hypothesis (unconditional for $|\mathcal{A}| = 2$ and the verified families, open in general). In Level 3 it determines the supertile hierarchy, yielding local witness injectivity. The remaining geometric step is reformulated as the SCC Producer Theorem: every recurrent noncoincident strongly connected component of $\mathcal{B}_\sigma$ produces a coincidence sibling at some inflation step. We give a mass-balance spectral characterization of this condition, prove that the closed-nonproductive case forces the within-SCC matrix to contain M_σ as an invariant sub-block, record an alphabet-3 spectral mechanism — any *closed* recurrent SCC with nonzero dominant wedge projection has within-SCC Perron eigenvalue at least $\beta|\beta_2|$, while the existence of such an SCC (the pure β-zero/escape alternative) is deferred to a companion β-richness analysis — and state the remaining open content. Pure discrete spectrum is then extracted via Barge–Štimac–Williams [12], the corrected one-dimensional bridge theorem.

1 Introduction

The Pisot substitution conjecture (PSC) asserts that the tiling dynamical system of a primitive irreducible Pisot substitution has pure discrete spectrum. Verified for two-letter alphabets by Hollander–Solomyak [16] and for various families by Barge–Diamond [10], Ito–Rao [17], Barge–Kwapisz [11], Akiyama–Lee [2], Barge [6], and others, the general case across all alphabet sizes has remained open; see the survey [1].

This paper revisits the conjecture in the standard formulation. The proof structure proceeds through two levels:

(1) *Level 2*: the balanced-pair automaton $\mathcal{B}_\sigma$ is finite.

(2) *Level 3*: finiteness implies that every balanced pair eventually reaches coincidence.

Our contribution is a structural simplification of Level 2 and a partial simplification of Level 3, with a single engine: the σ-images form a uniquely decodable code (Theorem 3.3), a short corollary of the classical defect theorem of Berstel–Perrin–Perrot–Restivo [13] combined with the linear algebra of the incidence matrix.

In Level 2, unique decodability rules out complete-cutting non-diagonal phase cycles through an explicit phase automaton, giving an effective constant $D(\sigma)$ (§4); finiteness of $\mathcal{B}_\sigma$ is carried as Hypothesis 4.10 (G1).

In Level 3, unique decodability for powers of σ determines the full hierarchy within every supertile and yields a local form of witness injectivity (§5). The remaining geometric step requires reformulation. The cycle-exclusion target ("no recurrent noncoincident cycle in $\mathcal{B}_\sigma$"), pursued in earlier drafts, is false as a general theorem: the flipped-Tribonacci substitution $\sigma : 1 \mapsto 21,\ 2 \mapsto 31,\ 3 \mapsto 1$ has a recurrent noncoincident cycle of period 3 with vanishing displacement. The correct target is the *SCC Producer Theorem*: every recurrent noncoincident strongly connected component of $\mathcal{B}_\sigma$ produces a coincidence sibling at some inflation step. Section 6 gives the mass-balance reformulation, the algebraic structure forced by the closed-nonproductive case, an alphabet-3 spectral lower bound on the within-SCC matrix, and the precise open content; the conjecture is essentially the Strong Coincidence Conjecture of Hollander–Solomyak [16] restated at the SCC level, settled in the two-letter case and open for $|\mathcal{A}| \geq 3$.

Earlier treatments invoked Mossé-style recognizability repeatedly throughout both levels. Here we isolate recognizability to a single role (the supertile hierarchy on the hull), while unique decodability drives the substantive arguments in Level 2 and the local part of Level 3.

Main theorem. *Let σ be a primitive Pisot substitution on a finite alphabet whose incidence matrix has irreducible characteristic polynomial, and suppose its balanced-pair automaton $\mathcal{B}_\sigma$ is finite* (G1). *Then, conditional on the SCC Producer Theorem (Conjecture 6.13), the associated tiling dynamical system has pure discrete spectrum.* Equivalently,

$$\boxed{\mathcal{B}_\sigma \text{ finite } + \text{ SCC Producer } \Longrightarrow \mathit{purediscretespectrum}.}$$

Finiteness of $\mathcal{B}_\sigma$ is unconditional for $|\mathcal{A}| = 2$ and for the verified families; for general $|\mathcal{A}| \geq 3$ it is the standing hypothesis G1 (see Hypothesis 4.10 and the following remark).

To extract pure discrete spectrum from termination of the balanced-pair algorithm we use the bridge theorem of Barge–Štimac–Williams [12] (one-dimensional case, Corollary 2 there); see §1.1 for the precise statement and a brief discussion of why this is the correct citation rather than the older Barge–Kwapisz form.

The strong-coincidence content of the SCC Producer Theorem (Conjecture 6.13) has been verified computationally on 6,009 irreducible Pisot specimens across alphabet sizes $k = 3, 4, 5$ (exhaustively for $k = 3$, $|\sigma(a)| \leq 3$), including 2,244 non-unimodular specimens, with zero failures; see Remark 7.4. The converse direction of Theorem 3.3 (that every non-UD specimen has $\det M_\sigma = 0$) was verified on all 59,319 substitutions with $|\mathcal{A}| = 3$, $|\sigma(a)| \leq 3$: of these, 7,491 are non-UD, and all of those have $\det M_\sigma = 0$, consistent with the proof.

1.1 Coincidence conditions and bridges

A reviewer familiar with the area knows that "coincidence" is overloaded. Several inequivalent conditions appear in the PSC literature, and they are not freely interchangeable.

- *Strong coincidence* (Arnoux–Ito; Barge–Diamond [10]): for each $i, j \in \mathcal{A}$ there exist k, n with $\sigma^n(i)$ and $\sigma^n(j)$ sharing the k-th letter and equal prefix Parikh vectors before it. Sufficient for measure-theoretic isomorphism to a domain exchange on the Rauzy fractal under additional hypotheses.

- *Overlap coincidence* (Solomyak; Akiyama–Lee [2]): every overlap reaches coincidence under inflation. Equivalent to pure discrete spectrum for self-affine tilings with the Meyer property.

- *Coincidence rank one* (Barge [5]): the maximal-equicontinuous-factor map is a.e. one-to-one. Equivalent to pure discrete spectrum for the $\mathbb{R}$-action under Pisot family hypotheses.

- *Balanced-pair-algorithm termination*: starting from a balanced pair (u, v), iterated inflation and irreducible factorization eventually produces a coincidence.

These conditions are not equivalent in full generality. Akiyama–Lee [3] are explicit: outside two-letter substitutions, the relation between strong coincidence and pure discrete spectrum is not completely understood. The strong-coincidence-implies-pure discrete spectrum folklore relies on additional hypotheses (trivial height group, Meyer property, control-point conditions) that are not part of our standing assumptions.

We do not invoke that folklore. The bridge we use is:

Theorem 1.1 (Barge–Štimac–Williams [12], Corollary 2)**.** *Let σ be a primitive Pisot substitution whose incidence matrix has irreducible characteristic polynomial. If there exist letters $a \neq b$ in $\mathcal{A}$ such that the balanced-pair algorithm for (ab, ba) terminates with coincidence, then the $\mathbb{R}$-action on Ω_σ has pure discrete spectrum.*

Remark 1.2. The earlier formulation of this bridge appears as Proposition 17.4 of Barge–Kwapisz [11]. Barge–Štimac–Williams [12, §1] note that the proof of [11, Theorem 16.3], from which Proposition 17.4 follows, contains a gap that is not easily plugged by the techniques of that paper, and recover the one-dimensional case as their Corollary 2. We cite [12] accordingly.

Remark 1.3 (Strength of what we prove versus what the bridge needs). Conditional on Conjecture 6.13, Proposition 6.14 shows that for every pair $a \neq b$ the BPA expansion of (ab, ba) contains a coincidence sibling at some iterate. Theorem 1.1 requires only that this hold for one well-chosen pair. The forward implication used here is exactly that direction; the converse (which would be needed for an iff) is not used and not claimed.

1.2 High-degree algebraic conjugates

Classical PSC verifications [16, 17, 11, 5, 6] handle the contracting conjugate space directly: tiles are placed in the contracting subspace, Rauzy fractals are constructed and shown to tile by a measure or topological-rigidity argument, and pure discrete spectrum is extracted from a tiling property. As alphabet size grows, the contracting space has dimension $|\mathcal{A}| - 1$ and the conjugate arithmetic becomes correspondingly delicate.

The proof here uses Pisot growth $\beta > 1$ only twice, both in geometric-bookkeeping roles:

(i) *Finiteness* (Hypothesis 4.10, G1): the balanced-pair automaton $\mathcal{B}_\sigma$ is finite. This is unconditional for $|\mathcal{A}| = 2$ and the verified families; for general $|\mathcal{A}| \geq 3$ it is a standing hypothesis. (The predecessor-contraction argument of earlier versions, which sought to derive finiteness from the padding bound, is withdrawn; see the remark after Hypothesis 4.10.)

(ii) *Supertile diameter growth* (used in connection with Conjecture 6.13): a level-n supertile has diameter at least $C\beta^n$, large enough to underwrite the geometric arguments around the SCC Producer Theorem in the literature.

Neither use enters the contracting conjugate space. No fractal is constructed; no algebraic conjugate of β is named. The contracting conjugates appear only inside the Mossé recognizability radius $R(\sigma)$, which we use as a black box.

This is not a claim that PSC can be proved without Pisot geometry: $\beta > 1$ is essential, and the Pisot character of β enters indirectly through Mossé recognizability. The narrower claim is that the proof avoids case-by-case estimates in the contracting conjugate space, and so avoids the explicit conjugate arithmetic that becomes painful for large $|\mathcal{A}|$.

1.3 Computational verifications

Balchin–Rust's Grout software [4] implements the Akiyama–Lee algorithm and the balanced-pair algorithm and searches for failures of strong coincidence, balanced-pair termination, and pure discrete spectrum. Their published searches found no counterexample among 12,573,955 irreducible Pisot substitutions on three letters and 15,001 on four letters. These are checking runs, not proofs: each individual substitution is verified by running its automata to termination.

The role of the present proof is complementary: it explains why such searches succeed. Any irreducible Pisot substitution forms a uniquely decodable code (Theorem 3.3); unique decodability rules out complete-cutting non-diagonal phase cycles via an explicit phase automaton (Observation 4.7); under the hypothesis that $\mathcal{B}_\sigma$ is finite (G1; unconditional for $|\mathcal{A}| = 2$ and the verified families), finiteness plus unique-hierarchy-within-supertiles plus the SCC Producer Theorem (Conjecture 6.13) gives that for every $a \neq b$ the BPA expansion of (ab, ba) contains a coincidence sibling at some iterate (Proposition 6.14), which is exactly the hypothesis of the bridge (Theorem 1.1). The self-contained structural reason for everything except finiteness and the SCC Producer Theorem exists at the level of $\det M_\sigma \neq 0$ and the defect theorem; no specimen-specific computation is needed.

1.4 Adjacent claimed proofs

Nakaishi's preprint [22] (arXiv:2401.07771v1, January 2024) claims a proof of "one version of Pisot conjecture" via weak mixing of the prefix-suffix automaton's subshift of finite type. As of writing, the preprint remains at v1 with no journal reference. An earlier preprint by the same author [21] on the same problem was withdrawn in September 2023 after six versions. Nakaishi's route proves a tiling property of the Rauzy fractal in the representation space, invoking weak mixing of the SFT to control overlap; this routes through Akiyama–Lee-style overlap-coincidence machinery in the contracting space. The present work routes around the contracting space entirely. The two arguments are independent.

Mercat–Akiyama [18] give a sufficient geometric condition for irreducible unit Pisot substitutions to have pure discrete spectrum and verify the conjecture for several families. Barge [6] establishes PSC for the β-substitution family, subsumed by the present result but proved by a route specific to that family. Berthé–Steiner–Thuswaldner and collaborators establish pure discrete spectrum for large families of S-adic shifts under generalized Pisot and geometric coincidence assumptions, in a setting (sequences of substitutions) distinct from ours.

1.5 Why dropping unimodularity is safe

Some formulations of PSC include a unimodularity hypothesis $|\det M_\sigma| = 1$ (e.g. [7]). We do not assume this. The reason is that the irreducibility of the characteristic polynomial does the work that unimodularity is sometimes used for: it forces $\mathbb{Z}$-independence of the tile lengths via a cyclic-vector argument (the orbit of any nonzero vector under $M_\sigma^\top$ spans $\mathbb{Q}^d$, since the minimal polynomial coincides with the irreducible characteristic polynomial). This rules out the homological-Pisot counterexample mechanism of Barge–Bruin–Jones–Sadun [7]: their Theorem 1 produces homological Pisot substitutions of every algebraic degree without pure discrete spectrum, but those examples are by construction not irreducible. The Coincidence Rank Conjecture [7] (that the coincidence rank divides a power of $|\det M_\sigma|$) remains open in the homological setting; it is consistent with everything in the present paper.

2 Background

Let $\mathcal{A}$ be a finite alphabet with $|\mathcal{A}| \geq 2$. A substitution $\sigma : \mathcal{A}^* \to \mathcal{A}^*$ maps each letter to a nonempty word. Its incidence matrix M_σ has $(M_\sigma)_{ij} = |\sigma(j)|_i$. The Parikh homomorphism $\pi : \mathcal{A}^* \to \mathbb{Z}^{\mathcal{A}}$ sends a word to its letter-count vector; it satisfies $\pi(\sigma(a)) = \mathrm{col}_a(M_\sigma)$.

Definition 2.1 (Standing hypotheses)**.** Throughout, σ satisfies:

(i) *primitive*: some power of M_σ has all entries strictly positive;

(ii) *aperiodic*: no fixed point of σ is eventually periodic (automatic for primitive substitutions on $|\mathcal{A}| \geq 2$ letters [23]);

(iii) *irreducible*: the characteristic polynomial of M_σ is irreducible over $\mathbb{Q}$ (in particular $\det M_\sigma \neq 0$);

(iv) *Pisot*: the Perron–Frobenius eigenvalue $\beta > 1$ is a Pisot number.

The left Perron eigenvector ℓ assigns positive tile lengths. The hull Ω_σ is the orbit closure of a fixed point of σ under the shift.

Lemma 2.2 ($\mathbb{Z}$-independence of tile lengths)**.** *Under Definition 2.1, the tile lengths $(\ell_a)_{a\in\mathcal{A}}$ are $\mathbb{Z}$-linearly independent.*

Proof. Suppose $n \in \mathbb{Z}^{\mathcal{A}} \setminus \{0\}$ satisfies $\langle n, \ell^\top \rangle = 0$. Since the characteristic polynomial of $M_\sigma^\top$ is irreducible of degree $d = |\mathcal{A}|$, every nonzero vector is cyclic for $M_\sigma^\top$: the orbit $n, (M_\sigma^\top)n, \ldots, (M_\sigma^\top)^{d-1}n$ spans $\mathbb{Q}^d$. But $\langle (M_\sigma^\top)^k n, \ell^\top \rangle = \beta^k \langle n, \ell^\top \rangle = 0$ for all k, so $\ell^\top \perp \mathbb{Q}^d$, forcing $\ell^\top = 0$, contradicting positivity. □

We do not use Lemma 2.2 on the critical path of the proof; it is recorded here because the irreducibility hypothesis enters Theorem 1.1 via this property in the underlying argument of [12].

Theorem 2.3 (Mossé recognizability [19, 20])**.** *For every primitive aperiodic substitution σ, there exists $R = R(\sigma) > 0$ such that the level-1 supertile boundaries of any point in Ω_σ are determined by the radius-R local tile content. Every tiling in Ω_σ therefore admits a unique supertile hierarchy at every level.*

Definition 2.4 (Balanced-pair automaton)**.** A *balanced pair* (u, v) is a pair of distinct words with $\pi(u) = \pi(v)$. Applying σ and splitting at coincidences produces children. The directed graph of reachable balanced pairs is $\mathcal{B}_\sigma$. The balanced-pair algorithm for (u, v) *terminates with coincidence* if iterated inflation and irreducible factorization eventually produces a coincidence pair (a, a).

Part I: Level 2 — Finiteness of $\mathcal{B}_\sigma$

3 Unique decodability

We recall the classical defect theorem.

Theorem 3.1 (Defect theorem [13, 14]). *Let $X \subset \mathcal{A}^+$ be a finite subset. Then X^* is contained in a unique minimal free submonoid $H^* \subseteq \mathcal{A}^*$, where $H \subset \mathcal{A}^+$ is a finite code satisfying $|H| \le |X|$. Moreover, $|H| = |X|$ if and only if X is itself a code.*

Lemma 3.2 (Letter injectivity). *If $\det M_\sigma \neq 0$, then σ is injective on single letters.*

Proof. $\sigma(a) = \sigma(b)$ gives identical columns of M_σ, contradicting $\det \neq 0$. □

Theorem 3.3 (Unique decodability). *If $\det M_\sigma \neq 0$, the code $X := \{\sigma(a) : a \in \mathcal{A}\}$ is uniquely decodable.*

Proof. By Lemma 3.2, $|X| = |\mathcal{A}|$. Suppose X is not a code. By Theorem 3.1, $X \subseteq H^*$ for some finite code $H \subset \mathcal{A}^+$ with $|H| \le |\mathcal{A}| - 1$.

For each $a \in \mathcal{A}$, write $\sigma(a) = h_1 h_2 \cdots h_{k_a}$ with $h_i \in H$. Apply the Parikh homomorphism:

$$\mathrm{col}_a(M_\sigma) = \pi(\sigma(a)) = \sum_i \pi(h_i) \in \mathbb{Z}\text{-span}\{\pi(h) : h \in H\}.$$

Every column of M_σ therefore lies in a sublattice of $\mathbb{Z}^{\mathcal{A}}$ of rank at most $|H| \le |\mathcal{A}| - 1$. Hence the $|\mathcal{A}|$ columns of M_σ are $\mathbb{Q}$-linearly dependent, giving $\mathrm{rank}(M_\sigma) \le |\mathcal{A}| - 1$ and $\det M_\sigma = 0$. Contradiction. □

Corollary 3.4 (UD for powers). *For every $r \ge 1$, the code $\{\sigma^r(a) : a \in \mathcal{A}\}$ is uniquely decodable.*

Proof. $\det M_{\sigma^r} = (\det M_\sigma)^r \neq 0$. Apply Theorem 3.3 to σ^r. □

Remark 3.5 (On the inputs to UD). Theorem 3.3 requires only $\det M_\sigma \neq 0$. Neither primitivity, aperiodicity, the Pisot condition, nor any recognizability result is used. This is the cleanest possible UD statement: an elementary linear-algebraic consequence of the classical defect theorem. The proof also yields, as a corollary, a simplification of Berthé–Steiner–Thuswaldner–Yassawi [15, Thm 3.1] for the rank-$|\mathcal{A}|$ case: their full recognizability theorem for morphisms of rank $|\mathcal{A}|$ implies UD, but the defect-theorem route gives UD directly in five lines without going through the rotational-conjugation machinery. [15] remains the essential reference for broader settings ($|\mathcal{A}| = 2$ with $\det = 0$; permutative morphisms).

4 Unique decodability, padding, and the finiteness hypothesis

This section records the unique-decodability and phase-automaton facts formerly used in the Level-2 finiteness attempt — (i) unique periodic cutting from UD (Theorem 4.1) and (ii) the finite phase-automaton padding diagnostic (Observation 4.7) — and then states finiteness of $\mathcal{B}_\sigma$ as Hypothesis 4.10 (G1). The predecessor-contraction argument of earlier versions is withdrawn (see the remark after Hypothesis 4.10).

Theorem 4.1 (Unique periodic cutting). *Let W be a finite word such that W^n is a legal factor of the substitution language for every $n \ge 1$. If W admits two complete σ-cuttings $\sigma(a) = W = \sigma(b)$ with $a, b \in \mathcal{A}^*$, then $a = b$.*

Proof. By Theorem 3.3, the code $\{\sigma(c) : c \in \mathcal{A}\}$ is uniquely decodable, so the factorization $W = \sigma(c_1)\sigma(c_2)\cdots\sigma(c_r)$ is unique. Hence $a = b$. ☐

Remark 4.2. By a *complete* σ-cutting of W we mean a factorization $W = \sigma(c_1)\cdots\sigma(c_r)$ in which the cuts align with σ-image boundaries throughout, with no boundary fragments. The padding analysis below converts a long coincidence run between the two sides of an inflated balanced pair into exactly this situation. Cuttings with proper-prefix or proper-suffix offset fragments are not used.

4.1 The phase automaton

To bound coincidence padding inside an inflated balanced pair, we track the local phase data position-by-position through a finite directed graph.

Definition 4.3 (Phase automaton)**.** The phase automaton Φ_σ has:

- Vertices: quadruples (α, i, β, j) with $\alpha, \beta \in \mathcal{A}$, $0 \le i < |\sigma(\alpha)|$, $0 \le j < |\sigma(\beta)|$, satisfying $\sigma(\alpha)_{i+1} = \sigma(\beta)_{j+1}$ (the next tile on each side coincides). Such a vertex is a *phase state.*
- Edges: a phase state advances by reading one tile on each side simultaneously. Within an image, $(\alpha, i, \beta, j) \to (\alpha, i+1, \beta, j+1)$ when $i + 1 < |\sigma(\alpha)|$ and $j + 1 < |\sigma(\beta)|$, provided the new vertex satisfies the coincidence condition. At the end of an image, $(\alpha, |\sigma(\alpha)| - 1, \beta, j) \to (\alpha', 0, \beta, j+1)$ with α' the next source letter on the u-side, and analogously when the v-side image ends; both ends terminating simultaneously corresponds to a transition to a fresh letter pair (α', β').

A phase state (α, i, β, j) is *diagonal* if $\alpha = \beta$ and $i = j$, otherwise *non-diagonal.* The number of phase states is at most

$$N_\Phi(\sigma) := |\mathcal{A}|^2 \cdot |\sigma|^2_{\max}, \qquad |\sigma|_{\max} := \max_{c \in \mathcal{A}} |\sigma(c)|.$$

A maximal coincidence run inside the inflated pair $(\sigma(u), \sigma(v))$ corresponds to a directed path in Φ_σ: each position of the run advances the phase state by one tile-step.

Lemma 4.4 (Long coincidence run forces a phase cycle)**.** *If a maximal coincidence run inside the inflation of a legal balanced pair has tile-length greater than $N_\Phi(\sigma)$, then the corresponding path in Φ_σ contains a directed cycle.*

Proof. A path of length exceeding the vertex count revisits a vertex, producing a cycle. ☐

Lemma 4.5 (A complete-cutting non-diagonal cycle gives two complete cuttings)**.** *Suppose Φ_σ contains a directed cycle γ realized inside a maximal coincidence run of an inflated legal balanced pair, in which at least one phase state is non-diagonal,* and γ returns to zero source-offset on both sides *(a complete-cutting cycle). Let W be the tile word emitted by γ over one period. Then there exist source words $a, b \in \mathcal{A}^*$ with $a \ne b$ such that $\sigma(a) = W = \sigma(b)$. If, moreover, γ is legally repeatable — so that W^n is a legal factor for every $n \ge 1$ — then unique periodic cutting (Theorem 4.1) is contradicted.*

Proof. Because γ returns to zero offset on each side, traversing γ once the u-side reads complete images: source letters $a_1, \ldots, a_{k_u}$ with $\sum_l |\sigma(a_l)| = |W|$ and no image split across the cycle seam; likewise the v-side reads $b_1, \ldots, b_{k_v}$ with $\sum_l |\sigma(b_l)| = |W|$. (The zero-offset hypothesis is exactly what makes these complete concatenations rather than mid-image fragments.) The cycle's defining property is that at every position the next tile on each side agrees, so $\sigma(a)$ and $\sigma(b)$ produce the same tile sequence W with $a = a_1 \cdots a_{k_u}$, $b = b_1 \cdots b_{k_v}$.

Suppose for contradiction $a = b$. Then the source sequences coincide letter-by-letter and the offsets evolve in lockstep, so every state on γ has equal offsets and equal source letters, i.e. is diagonal — contradicting the non-diagonal hypothesis. Hence $a \neq b$.

For the final clause, if W^n is a legal factor for every n, then W admits the two complete σ-cuttings $\sigma(a) = W = \sigma(b)$ with $a \neq b$ in the presence of unbounded legal repetition, which Theorem 4.1 forbids. We do not assert that an arbitrary complete-cutting cycle is legally repeatable; the contradiction is available precisely in the periodic-realizable case. □

Remark 4.6 (Non-complete cycles are not controlled). A phase cycle need not return to zero offset: it may recur at an offset pair $(\alpha, \beta) \neq (0, 0)$, emitting a cyclic fragment that is a mid-image piece rather than a complete concatenation $\sigma(a) = W = \sigma(b)$. Direct enumeration finds such non-complete recurrences at a nonzero offset in roughly one fifth of recurring phases on the alphabet-3 corpus. For these the unique-decodability contradiction does not apply, so Lemma 4.5 controls only complete-cutting cycles. This is why the padding statement below is recorded as a diagnostic observation rather than a theorem.

Observation 4.7 (Padding diagnostic, non-diagonal phase). The phase automaton Φ_σ is finite, with at most $|\mathcal{A}|^2 \cdot |\sigma|^2_{\max}$ states; set $D(\sigma) := |\mathcal{A}|^2 |\sigma|^2_{\max}$. A maximal coincidence run whose non-diagonal phase path contains a *legally repeatable complete-cutting* cycle cannot occur in a legal balanced pair, since it would force $\sigma(a) = \sigma(b)$ with $a \neq b$ under unbounded legal repetition, contradicting Theorem 4.1 (Lemma 4.5). This excludes the repeatable complete-cutting recurrences. Non-complete (nonzero-offset) recurrences, and complete-cutting cycles not known to be legally repeatable, are not controlled by this argument (Remark 4.6); a uniform bound on the total non-diagonal phase length is therefore recorded here as a diagnostic, not proved.

Remark 4.8 (Scope of the diagnostic). This is not used elsewhere in the paper: finiteness of $\mathcal{B}_\sigma$ is the hypothesis G1, not a consequence of padding. We record the phase-automaton finiteness and the complete-cutting obstruction because they explain *why* long coincidence runs are atypical, and because the non-diagonal phase length is the natural obstruction quantity; but the only rigorous content is the complete-cutting exclusion via Theorem 4.1. Direct computation exhibits maximal coincidence runs far longer than $D(\sigma)$ (total length up to several multiples on the alphabet-3 corpus), the excess lying in diagonal phase and in nonzero-offset non-diagonal recurrence; the earlier "every maximal coincidence run has tile-length at most $D(\sigma)$" is withdrawn.

Remark 4.9 (Comparison with v6.2 and v10). Earlier formulations (v6.2 §§3–4; v10 Corollary 4.2) sketched this argument under the heading "non-diagonal phase cycle contradicts unique periodic cutting" without defining the phase automaton or quantifying D. Observation 4.7 makes both precise; in the present version this constant $D(\sigma) \leq |\mathcal{A}|^2 |\sigma|^2_{\max}$ is recorded as a local diagnostic consequence of unique decodability, not used to prove finiteness (which is carried as Hypothesis 4.10, G1).

4.2 Predecessor contraction

Hypothesis 4.10 (Finiteness, G1)**.** *$\mathcal{B}_\sigma$ is finite.*

Remark 4.11 (Status of Hypothesis 4.10). Finiteness is taken as a hypothesis, not proved here. Earlier versions (through v14) derived it from a *predecessor contraction*: with $D = D(\sigma)$ from Observation 4.7 and $L(s)$ the geometric length under the Perron tile-length vector, it was claimed that any child s of a parent s' satisfies $L(s) \geq \beta L(s') - D \cdot \ell_{\max}$, whence $L(s') \leq (L(s) + D \cdot$

$\ell_{\max})/\beta$ and backward iteration contracts at rate $1/\beta$. *This step is false.* The padding bound of Observation 4.7 bounds the geometric length of each removed coincidence run, but inflation of s' can split into many noncoincident children, and a chosen child may carry only a small fraction of the parent's inflated length while the remaining noncoincident mass lies in other children. Thus no single child need carry almost all of $\beta L(s')$, and the inequality $L(s) \geq \beta L(s') - D \cdot \ell_{\max}$ does not hold; direct computation exhibits children with $\beta L(s)/L(s')$ unbounded (ratios up to 8 and growing with state length on the alphabet-3 corpus). The padding bound controls individual coincidence runs, not total deletion mass, and the contraction it was meant to support fails.

Finiteness is known unconditionally for $|\mathcal{A}| = 2$ (Hollander–Solomyak with Barge–Diamond) and for the verified families (beta-substitutions, Brun/Jacobi–Perron products, and assorted Ito–Rao and Akiyama–Lee cases). For general irreducible Pisot σ on $|\mathcal{A}| \geq 3$ it is open; the exhaustive alphabet-3 balanced-pair sweep (4,554/4,554 specimens close, zero cap hits) is empirical elimination, not a proof. We therefore carry finiteness as the hypothesis G1 throughout.

Part II: Level 3 — Coincidence

5 Unique hierarchy within supertiles

Lemma 5.1 (Unique hierarchy)**.** *Let T be a level-n supertile of type τ with tile content $V = \sigma^n(\tau)$. For each $1 \leq k \leq n$, the level-k supertile boundaries within T are uniquely determined by V.*

Proof. $V = \sigma^k(w)$ with $w = \sigma^{n-k}(\tau)$, so V is a concatenation of σ^k-images starting and ending at level-n boundaries (hence level-k boundaries). By Corollary 3.4, the σ^k-parsing is unique. Recursive application at each level gives the full hierarchy. □

Definition 5.2 (Pointed realization, witness, supertile window)**.** A *pointed realization* is a triple (ω, ω', x_0) with $(\omega, \omega') \in \Omega_\sigma \times \Omega_\sigma$ disagreeing only on a finite disagreement window and $x_0 \in \mathbb{R}$ a marked point in that window. The *centered state* $s(x_0)$ records the local balanced pair and the position of x_0 within it. The *level-n ancestry* $a_n(x_0)$ records the type of the level-n supertile containing x_0 in ω, together with x_0's position within that supertile. The *finite witness* is $W_n(x_0) := (s(x_0), a_n(x_0))$.

The *level-n supertile window* of a pointed realization is the restriction of (ω, ω') to the level-n supertile of ω containing x_0, considered as a pair of partial tilings together with x_0.

Theorem 5.3 (Local witness injectivity)**.** *Let $(\omega_1, \omega_1', x_0)$ and $(\omega_2, \omega_2', x_0)$ be two pointed realizations sharing centered state s and level-n ancestry a_n. Then their level-n supertile windows agree: ω_1 and ω_2 coincide on the level-n supertile of ω_1 containing x_0, with identical level-k hierarchical structure for every $k \leq n$, and analogously for ω_1' and ω_2'.*

Proof. The shared level-n ancestry places x_0 inside the same level-n supertile T of type τ in both realizations, at the same position within T. The tile content of T is $V = \sigma^n(\tau)$ in both. By Lemma 5.1, the level-k supertile boundaries within V are determined by V alone for every $k \leq n$, hence agree across the two realizations. Therefore ω_1 and ω_2 coincide on T together with their hierarchies. The shared centered state gives the same local balanced-pair structure at x_0, hence the same disagreement-window data for ω_1' and ω_2' within T. □

Remark 5.4 (On the strength of Theorem 5.3)*.* Earlier formulations (v9, v10) stated witness injectivity globally: same finite witness W_n for all n implies same pointed realization. That stronger conclusion does not follow from finite data, since W_n records only local content within a single level-n supertile and cannot determine an entire bi-infinite tiling. The local form stated here is what is needed downstream and what the proof actually establishes; v6.2 stated this form correctly, and v9–v10 compressed it incorrectly.

6 Coincidence

The remaining step is to show that finiteness of $\mathcal{B}_\sigma$ implies every balanced pair eventually reaches a coincidence state. The natural reformulation in the language of strongly connected components (SCCs) of $\mathcal{B}_\sigma$ is:

> *Every recurrent noncoincident SCC of $\mathcal{B}_\sigma$ produces a coincidence sibling at some inflation step.*

Earlier formulations (including v9, v10, and v11 prior to revision) attempted to prove the stronger statement that no recurrent noncoincident cycle exists in $\mathcal{B}_\sigma$, with the substantive geometric content being nonvanishing of cycle displacement $\delta(\gamma) \neq 0$ on noncoincident cycles. That target is false as a general theorem. We discuss what fails, give a sharper algebraic reformulation (the mass-leakage framework), record an alphabet-3 spectral lower bound, and identify the precise open content.

6.1 Why cycle exclusion fails

The flipped Tribonacci substitution $\sigma : 1 \mapsto 21,\ 2 \mapsto 31,\ 3 \mapsto 1$ is primitive Pisot with irreducible characteristic polynomial $x^3 - x^2 - x - 1$. Its balanced-pair automaton contains a recurrent SCC of size four,

$$\mathcal{C}_{\mathrm{fT}} = \{((1,2),(2,1)),((1,3),(3,1)),((2,1,3),(3,1,2)),((1,2,1,3),(3,1,2,1))\},$$

which is closed (no escape) and contains a directed cycle of period 3 on the first three states. The cycle takes the leftmost child at every inflation step, so every offset is 0 and the cycle displacement vanishes,

$$\delta(\gamma) = 0.$$

Hence "noncoincident cycle $\Rightarrow \delta(\gamma) \neq 0$" is false. Independently, a Mossé desubstitution of the cycle's asymptotic pair returns into the same SCC rather than to a smaller balanced-pair state, so the σ^q-self-similarity of the cycle blocks any descent argument that aims to produce a strictly smaller noncoincident state.

The correct mechanism is not exclusion. The same SCC produces coincidence siblings at every step: each application of σ to a state $s \in \mathcal{C}_{\mathrm{fT}}$ generates one single-tile coincidence child of type $(1)|(1)$ alongside the within-SCC noncoincident children. The balanced-pair algorithm terminates because of this coincidence production, not because the SCC is destroyed.

6.2 Mass balance and the spectral characterization

Fix σ primitive Pisot with irreducible characteristic polynomial. For a balanced pair $s = (u, v)$ write $\pi(s) := \pi(u) = \pi(v) \in \mathbb{Z}_{\geq 0}^{\mathcal{A}}$ for the common Parikh vector and $L(s) := \langle \ell, \pi(s) \rangle$ for its ℓ-length, where ℓ is the left Perron eigenvector of M_σ with $\ell M_\sigma = \beta \ell$. Inflation gives $L(\sigma(s)) = \beta L(s)$, and irreducible factorization of $\sigma(s)$ at coincidence boundaries decomposes $\sigma(s)$ into noncoincident children and single-tile coincidence siblings.

Let $\mathcal{C} \subseteq \mathcal{B}_\sigma$ be a noncoincident SCC. Define the within-SCC transition matrix

$$N_\mathcal{C} \in \mathrm{Mat}_{|\mathcal{C}|}(\mathbb{Z}_{\geq 0}), \qquad N_\mathcal{C}[s,t] = \#\{\text{children of } \sigma(s) \text{ of type } t\}, \quad s,t \in \mathcal{C}.$$

Mass balance at each $s \in \mathcal{C}$ reads

$$\beta L(s) = \sum_{t \in \mathcal{C}} N_\mathcal{C}[s,t]\, L(t) + \mathrm{escape}(s) + \mathrm{leak}(s), \tag{1}$$

where escape(s) collects L-mass of noncoincident children outside $\mathcal{C}$ and leak(s) collects L-mass of coincidence (diagonal) children. We say $\mathcal{C}$ is *closed* if every noncoincident child of every $s \in \mathcal{C}$ lies in $\mathcal{C}$, equivalently escape $\equiv 0$, and *nonproductive* if no $\sigma(s)$ for $s \in \mathcal{C}$ has a single-tile coincidence child, equivalently leak $\equiv 0$.

Lemma 6.1 (Mass-balance spectral characterization)**.** *Let $\mathcal{C}$ be a recurrent noncoincident SCC. Then $L_\mathcal{C} := (L(s))_{s \in \mathcal{C}} \in \mathbb{R}^{|\mathcal{C}|}_{>0}$ is a positive vector, and*

$$N_\mathcal{C} L_\mathcal{C} \leq \beta L_\mathcal{C} \quad \textit{componentwise.}$$

Moreover the following are equivalent:

(i) $\mathcal{C}$ is closed and nonproductive (i.e., escape $\equiv$ leak $\equiv 0$*);*

(ii) $N_\mathcal{C} L_\mathcal{C} = \beta L_\mathcal{C}$;

(iii) the Perron–Frobenius eigenvalue of $N_\mathcal{C}$ equals β.

If any of these fails, then $\mathrm{PF}(N_\mathcal{C}) < \beta$ *strictly.*

Proof. Mass balance (1) gives $N_\mathcal{C} L_\mathcal{C} \leq \beta L_\mathcal{C}$. Equality of (i) and (ii) is immediate from (1). The implication (ii)⇒(iii) follows from the existence of a positive eigenvector with eigenvalue β, which is then the spectral radius by Perron–Frobenius for nonnegative matrices. For (iii)⇒(ii): since $\mathcal{C}$ is recurrent, $N_\mathcal{C}$ is irreducible; if $\mathrm{PF}(N_\mathcal{C}) = \beta$ then by Perron–Frobenius the right β-eigenspace is one-dimensional and spanned by a positive vector, which must be $L_\mathcal{C}$ up to scalar, forcing equality in (1). The strict inequality in the failure case follows from the Collatz–Wielandt characterization of $\mathrm{PF}(N_\mathcal{C})$. □

6.3 Algebraic structure of a closed nonproductive SCC

Suppose $\mathcal{C}$ is closed nonproductive. Let $P \in \mathrm{Mat}_{|\mathcal{C}| \times |\mathcal{A}|}(\mathbb{Z}_{\geq 0})$ be the matrix whose rows are the Parikh vectors $\pi(s)^\top$ for $s \in \mathcal{C}$. Since $\pi(\sigma(s)) = M_\sigma \pi(s)$ and the children's Parikhs sum to $\pi(\sigma(s))$ with no coincidence Parikh, we have, in matrix form,

$$N_\mathcal{C} P = P\, M_\sigma^\top. \tag{2}$$

Lemma 6.2 (Algebraic structure)**.** *If $\mathcal{C}$ is closed nonproductive, then in addition to* (2)*,*

(i) $\ker(P : \mathbb{R}^{|\mathcal{A}|} \to \mathbb{R}^{|\mathcal{C}|})$ *is $M_\sigma^\top$-invariant and defined over $\mathbb{Q}$;*

(ii) by irreducibility of the characteristic polynomial of M_σ over $\mathbb{Q}$, $\ker(P) \in \{0, \mathbb{R}^{|\mathcal{A}|}\}$*, and since $\pi(s) \neq 0$ for all $s \in \mathcal{C}$,* $\ker(P) = 0$*. Hence P is injective, and $|\mathcal{C}| \geq |\mathcal{A}|$;*

(iii) $N_\mathcal{C}$ contains M_σ as a sub-block on $\mathrm{im}(P)$*: the eigenvalues of $N_\mathcal{C}$ restricted to* $\mathrm{im}(P)$ *are precisely the eigenvalues of M_σ, namely β together with its $|\mathcal{A}|-1$ Galois conjugates (all with $|\cdot| < 1$ by the Pisot property).*

Proof. For (i): if $Py = 0$ then $P(M_\sigma^\top y) = N_\mathcal{C}(Py) = 0$ by (2), so $\ker(P)$ is $M_\sigma^\top$-invariant; it is defined over $\mathbb{Q}$ because P has integer entries. For (ii): the only $M_\sigma^\top$-invariant $\mathbb{Q}$-subspaces of $\mathbb{R}^{|\mathcal{A}|}$ are $\{0\}$ and $\mathbb{R}^{|\mathcal{A}|}$, by irreducibility of $\mathrm{char}(M_\sigma)$ over $\mathbb{Q}$; since P is not the zero matrix, $\ker(P) = 0$. Then $|\mathcal{C}| = \mathrm{rank}(P) \geq \dim(\mathbb{R}^{|\mathcal{A}|}) = |\mathcal{A}|$. For (iii): P intertwines $M_\sigma^\top$ on $\mathbb{R}^{|\mathcal{A}|}$ with $N_\mathcal{C}|_{\mathrm{im}(P)}$, so the spectra match. $\square$

6.4 Spectral lower bound on $N_\mathcal{C}$ (alphabet 3)

For alphabet size three, an additional spectral lower bound on the within-SCC matrix is available. The construction tracks the second antisymmetric tensor power of the Parikh count and is alphabet-3-specific: the dimension count $\binom{3}{2} = 3$ plus the standard-basis identity $K_2(s_{ab}) = e_{ab}$ on the three length-2 swap pairs make the projection onto the dominant eigenline of $\Lambda^2 M_\sigma$ verifiable in finite cases.

For $|\mathcal{A}| = 3$, write $\Lambda^2 M_\sigma$ for the induced action on $\Lambda^2\mathbb{Z}^3 \cong \mathbb{Z}^3$, with eigenvalues $\beta\beta_2$, $\beta\beta_3$, $\beta_2\beta_3$. The Pisot ordering $\beta > 1 > |\beta_2| \geq |\beta_3|$ gives $|\beta\beta_2| > |\beta\beta_3|$ and $|\beta\beta_2| \geq |\beta_2\beta_3|$, so $|\beta\beta_2|$ is the dominant eigenvalue magnitude of $\Lambda^2 M_\sigma$. Let $\Pi^\sigma_{\mathrm{dom}}$ denote the spectral projection of $\Lambda^2 M_\sigma$ onto its dominant eigenspace.

For a balanced pair $s = (u, v)$ with $|\mathcal{A}| = 3$, define

$$K_2(s) := (N_{ij}(u) - N_{ij}(v))_{1\leq i<j\leq 3} \in \mathbb{Z}^3,$$

where $N_{ij}(w)$ counts ordered i-before-j scattered pairs in w.

Lemma 6.3 (Dominant K_2-source). *For every primitive irreducible Pisot substitution σ on $\mathcal{A} = \{1,2,3\}$, at least one of the three length-2 swap pairs $s_{12} = ((1,2),(2,1))$, $s_{13} = ((1,3),(3,1))$, $s_{23} = ((2,3),(3,2))$ satisfies $\Pi^\sigma_{\mathrm{dom}} K_2(s_0) \neq 0$.*

Proof sketch. Direct computation in the (e_{12}, e_{13}, e_{23}) basis of $\Lambda^2\mathbb{Z}^3$ gives $K_2(s_{12}) = e_{12}$, $K_2(s_{13}) = e_{13}$, $K_2(s_{23}) = e_{23}$. These are the standard basis of $\Lambda^2\mathbb{Z}^3$. If $\Pi^\sigma_{\mathrm{dom}}$ vanished on all three, it would vanish on $\Lambda^2\mathbb{R}^3$, contradicting that the dominant eigenspace is nonzero (which holds because $\beta\beta_2 \neq 0$ for $\det M_\sigma \neq 0$). Hence some s_0 has nonzero dominant projection. $\square$

Proposition 6.4 (Spectral lower bound on $N_\mathcal{C}$, closed SCC, alphabet 3)**.** *Let σ be a primitive Pisot substitution on $\mathcal{A} = \{1,2,3\}$ with irreducible characteristic polynomial, and suppose $\mathcal{B}_\sigma$ is finite* (G1)*. For any* closed *recurrent noncoincident SCC $\mathcal{C}$ with nonvanishing dominant wedge component $\Pi^\sigma_{\mathrm{dom}} K_2|_\mathcal{C} \neq 0$,*

$$\mathrm{PF}(N_\mathcal{C}) \geq \beta\,|\beta_2|.$$

Proof sketch. By the K_2-pushforward identity $K_2(\sigma s) = \Lambda^2 M_\sigma \cdot K_2(s)$ on whole pairs (companion note), for $s \in \mathcal{C}$ with $\Pi^\sigma_{\mathrm{dom}} K_2(s) \neq 0$, $\|\Lambda^2 M_\sigma^k\, K_2(s)\| \gtrsim (\beta|\beta_2|)^k$. An inter-block cancellation argument shows $K_2(\sigma^k s) = \sum_{c\in D_k} K_2(c)$, with D_k the depth-k descendants; the triangle inequality forces $\sum_{c\in D_k} \|K_2(c)\| \gtrsim (\beta|\beta_2|)^k$. *Because $\mathcal{C}$ is closed*, every descendant of $s \in \mathcal{C}$ remains in $\mathcal{C}$ (noncoincident children stay in $\mathcal{C}$; coincidence children carry no wedge mass), so $D_k \subseteq \mathcal{C}$ and the wedge growth cannot escape into other components. By Hypothesis 4.10, $\|K_2(c)\|$ is uniformly bounded, so $|D_k| \gtrsim (\beta|\beta_2|)^k$; path-counting within $\mathcal{C}$ gives $|D_k| \leq C\,\mathrm{PF}(N_\mathcal{C})^k$, whence $\mathrm{PF}(N_\mathcal{C}) \geq \beta|\beta_2|$. Full proof in the companion note (§7.5). The closedness hypothesis is essential: for an open SCC, descendants can leave $\mathcal{C}$ and the dominant growth may be carried by escaping components, so $N_\mathcal{C}$ need not see it. $\square$

Observation 6.5 (Conditional existence of a qualifying SCC). Starting from a dominant length-2 seed $s_0 \in \{s_{12}, s_{13}, s_{23}\}$ with $\Pi^{\sigma}_{\text{dom}} K_2(s_0) \neq 0$ (Lemma 6.3), the global descendant growth forces some recurrent component of the condensation graph to carry the spectral growth $(\beta|\beta_2|)^k$, *provided* the dominant wedge mass is not lost through escape into pure β-zero components ($\Pi^{\sigma}_{\text{dom}} K_2 \equiv 0$). Excluding that alternative is conditional on one of two auxiliary lemmas (No Pure β-Zero SCC, or Case-B β-richness/cyclic-vector) recorded in the companion note. The implication of Proposition 6.4 is theorem-grade for a closed SCC on alphabet 3; the existence of a qualifying (closed, wedge-nonzero) SCC is this conditional part.

Remark 6.6 (What Proposition 6.4 contributes and what it does not). Proposition 6.4 is alphabet-3 specific and gives a lower bound on $\mathrm{PF}(N_{\mathcal{C}})$ for any *closed* recurrent SCC with nonzero dominant wedge (existence of such an SCC being conditional, Observation 6.5). It does not close the SCC Producer Theorem (Conjecture 6.13), even on alphabet 3. By Lemma 6.1(iii), the closed-nonproductive obstruction corresponds to $\mathrm{PF}(N_{\mathcal{C}}) = \beta$, which is consistent with $\mathrm{PF}(N_{\mathcal{C}}) \geq \beta|\beta_2|$ (since $|\beta_2| < 1$ gives $\beta|\beta_2| < \beta$). Proposition 6.4 therefore constrains the bottom of the spectral band $[\beta|\beta_2|, \beta]$ but not the top, where the obstruction lives. Its value is rather:

- it rules out such spectral collapse for any closed recurrent SCC with nonzero dominant wedge projection;
- it confirms the framework of §6.2–§6.4 is consistent and substantive on alphabet 3;
- it gives a quantitative cardinality estimate $|D_k| \gtrsim (\beta|\beta_2|)^k$ for descendants of a length-2 dominant seed.

6.5 The cohomological face of the obstruction: an independent development

The closed-nonproductive SCC of §6.2–§6.4 admits a second, independent description on the cohomology side of the Barge–Diamond theory [8, 9], developed in a companion manuscript. The two descriptions concern the *same* object: the intertwining $N_{\mathcal{C}} P = P\, M_\sigma^\top$ of (2) with P injective (Lemma 6.2) is exactly the statement that the integer valuation P embeds $\mathrm{spec}(M_\sigma)$ into $\mathrm{spec}(N_{\mathcal{C}})$, i.e. $\chi_{M_\sigma} \mid \chi_{N_{\mathcal{C}}}$, with $\mathrm{PF}(N_{\mathcal{C}}) = \beta$ (Lemma 6.1(iii)). On the cohomology side this $N_{\mathcal{C}}$ is the abelianization of a substitution $\Sigma_{\mathcal{C}}$ on $|\mathcal{C}| \geq |\mathcal{A}|$ letters—the *transversal substitution* of the trapped component—whose self-similar tiling space $\Omega_{\Sigma_{\mathcal{C}}}$ has dilatation the Pisot β and reducible characteristic polynomial $\chi_{M_\sigma} \cdot q$. The matrix $N_{\mathcal{C}}$ is irreducible ($\mathcal{C}$ recurrent) but need not be primitive; after passing to a suitable power, primitivity of the induced component can be arranged; irreducibility of the powered incidence matrix is not needed for the local cohomological discussion here, and primitivity of $\Sigma_{\mathcal{C}}$ is assumed throughout this subsection.

This reframing does not close Conjecture 6.13, but it contributes three facts—two unconditional and one a second conditional route—that constrain the hypothetical trapped object from a direction disjoint from the spectral lower bound of Proposition 6.4.

Lemma 6.7 (Leg factor). *If $\mathcal{C}$ is a closed nonproductive SCC, the tiling flow $(\Omega_{\Sigma_{\mathcal{C}}}, \mathbb{R})$ factors onto $(\Omega_\sigma, \mathbb{R})$.*

The factor map merges the cells of a $\Sigma_{\mathcal{C}}$-tiling between consecutive boundaries of one of the two underlying σ-grids; the boundary types are intrinsic (one letter changes across each), the merge is a local rule, and its image is a nonempty closed translation-invariant subset of the minimal Ω_σ, hence all of it. Pure discrete spectrum passes to topological factors. To conclude that $\Omega_{\Sigma_{\mathcal{C}}}$ lacks pure discrete spectrum one needs that a closed nonproductive SCC certifies $\neg$*purediscretespectrum* for σ — equivalently, that balanced-pair non-coincidence implies overlap non-coincidence, the *converse* direction of the bridge, which Theorem 1.1 does not supply but which is available through the Akiyama–Lee overlap-coincidence criterion (*purediscretespectrum* $\Leftrightarrow$ overlap coincidence for irreducible Pisot). Under that converse:

Corollary 6.8 (Coincidence-rank floor, conditional on the bridge converse)**.** *Assume the converse direction above (a closed nonproductive SCC certifies $\neg$purediscretespectrum for σ, via the Akiyama–Lee overlap-coincidence criterion). Then for every closed nonproductive SCC, $\Omega_{\Sigma_{\mathcal{C}}}$ does not have pure discrete spectrum; equivalently its coincidence rank satisfies* $\mathrm{cr}(\Sigma_{\mathcal{C}}) \geq 2$*. If in addition σ is unimodular and $\Sigma_{\mathcal{C}}$ is homological Pisot, then* $\mathrm{cr}(\Sigma_{\mathcal{C}}) \geq 3$*, by the odd-norm exclusion of* $\mathrm{cr} = 2$ *for homological Pisot substitutions* [5].

The triple structure forced by $\mathrm{cr} \geq 3$ obeys the same mass-balance ceiling as the pair structure:

Lemma 6.9 (Triple mass-balance ceiling)**.** *Let a closed nonproductive SCC carry, via* $\mathrm{cr} \geq 3$*, a system of triple-overlap states with the partition identity $\beta\, w(s) = \sum_c w(c)$ over the common refinement of the three inflated images. Then the within-system count matrix $N_{\mathcal{C}}^{(3)}$ has* $\mathrm{PF}(N_{\mathcal{C}}^{(3)}) = \beta$ *when the system is closed, and* $\mathrm{PF} < \beta$ *strictly otherwise—verbatim the dichotomy of Lemma 6.1.*

The proof is the Perron–Frobenius argument of Lemma 6.1 applied to the positive eigenvector $(w(s))_s$; the only new input, the triple partition identity, is the interval-partition identity for the common refinement of three grids rather than two. (A computational surrogate over producing components—$\sim$ 1,750 triple noncoincident SCCs across the alphabet-3 corpus—found the ceiling $\mathrm{PF}(N_{\mathcal{C}}^{(3)}) < \beta$ in every case with no closed triple system, mirroring the pair census; this is elimination, not validation.)

Proposition 6.10 (Second conditional route, unimodular case)**.** *Let σ be unimodular. Suppose (Q-A) the transversal substitution $\Sigma_{\mathcal{C}}$ of any closed nonproductive SCC is homological Pisot, and suppose the Coincidence Rank Conjecture* [7] *holds. Then σ has pure discrete spectrum; equivalently no closed nonproductive SCC exists.*

Proof. Suppose σ has a closed nonproductive SCC. Its dilatation β has norm ± 1 (unimodularity). Under (Q-A) $\Sigma_{\mathcal{C}}$ is homological Pisot; the Coincidence Rank Conjecture then forces $\mathrm{cr}(\Sigma_{\mathcal{C}})$ to divide a power of 1, i.e. $\mathrm{cr}(\Sigma_{\mathcal{C}}) = 1$, so $\Omega_{\Sigma_{\mathcal{C}}}$ has pure discrete spectrum. By Lemma 6.7 pure discrete spectrum passes to the factor Ω_σ, contradicting Corollary 6.8. □

Remark 6.11 (Relation to Proposition 6.4 and to Conjecture 6.13). Proposition 6.10 is a placement, not a closure: both (Q-A)—homological-Pisotness of $\Sigma_{\mathcal{C}}$, equivalently $\dim \check{H}^1(\Omega_{\Sigma_{\mathcal{C}}}) = \deg \beta$—and the Coincidence Rank Conjecture are open, the latter precisely in the $\mathrm{cr} \geq 3$ regime this route reaches. Its content is that the symbolic obstruction of Conjecture 6.13 and the topological homological-Pisot/CRC program are two faces of one object. Where Proposition 6.4 constrains the bottom of the spectral band $[\beta|\beta_2|, \beta]$, condition (Q-A) constrains the residual spectrum $q = \chi_{N_{\mathcal{C}}}/\chi_{M_\sigma}$ at the top: it would force every nonzero root of q onto a root of unity. The facts of Lemma 6.7 and Lemma 6.9 hold unconditionally; the coincidence-rank floor (Corollary 6.8) holds under the bridge converse (Akiyama–Lee overlap coincidence). These are recorded as the geometric counterparts of the algebraic structure of Lemma 6.2. No census evidence bears on either route, since a closed nonproductive SCC—the common object of both—occurs nowhere in the certified corpus.

6.6 The remaining open question

The combinatorial content of "$\mathcal{C}$ is closed nonproductive" is sharper than the algebraic constraints of Lemma 6.2: it requires that for every $s = (u, v) \in \mathcal{C}$ and every coincidence boundary k in the inflated pair $(\sigma(u), \sigma(v))$, the letters $\sigma(u)_k$ and $\sigma(v)_k$ are distinct. Equivalently, the prefix-difference walk $\Delta_s(k) := \pi(\sigma(u)_{[0,k)}) - \pi(\sigma(v)_{[0,k)}) \in \mathbb{Z}^{|\mathcal{A}|}$ never returns to the origin at a position followed by a diagonal letter pair.

Definition 6.12. A *diagonal-free zero-return system* on $\mathcal{C}$ is the data $(\mathcal{C}, N_{\mathcal{C}}, P)$ satisfying (2) and the combinatorial constraint above (no zero-return of Δ_s followed by a diagonal letter pair, for any $s \in \mathcal{C}$).

Conjecture 6.13 (SCC Producer Theorem)**.** *For every primitive Pisot substitution σ with irreducible characteristic polynomial, no diagonal-free zero-return system exists. Equivalently, every recurrent noncoincident SCC of $\mathcal{B}_\sigma$ is productive: some $s \in \mathcal{C}$ has at least one coincidence sibling in $\sigma(s)$.*

The forward direction of the corresponding equivalence is what the proof of pure discrete spectrum requires:

Proposition 6.14. *Suppose $\mathcal{B}_\sigma$ is finite and Conjecture 6.13 holds. Then every recurrent noncoincident SCC of $\mathcal{B}_\sigma$ has a reachable state s such that $\sigma(s)$ contains a coincidence sibling. Hence every seed pair whose forward orbit in $\mathcal{B}_\sigma$ enters such an SCC has an iterate σ^m whose irreducible factorization contains a coincidence sibling; in particular this holds for the seed (ab, ba) for every pair $a \neq b$ in $\mathcal{A}$.*

Proof. The orbit of any seed in $\mathcal{B}_\sigma$, being a forward trajectory in a finite digraph, eventually enters a recurrent SCC $\mathcal{C}$. If $\mathcal{C}$ is coincident (contains a diagonal state (a, a)) the orbit already contains a coincidence sibling. Otherwise $\mathcal{C}$ is recurrent noncoincident; by Conjecture 6.13 there is a state $s_* \in \mathcal{C}$ with a coincidence sibling in $\sigma(s_*)$, and by strong connectedness of $\mathcal{C}$ every $s \in \mathcal{C}$ reaches s_* in finitely many steps. Composing with the prior segment of the orbit, the iterate σ^m for m large enough has an irreducible factorization containing a coincidence sibling. $\square$

6.7 Status and what is known

Conjecture 6.13 is essentially the Strong Coincidence Conjecture restated at the SCC level: it would follow from the (open) statement that for primitive Pisot σ with irreducible characteristic polynomial every noncoincident balanced pair has a productive iterate. What is known points in the right direction without closing the conjecture in general:

- Barge–Diamond [10] prove that for some letter pair $a \neq b$, $\sigma^n(a)$ and $\sigma^n(b)$ share a letter at the same position with matching prefix Parikhs at some $n \geq 0$. This gives productivity for one specific seed but does not propagate to arbitrary recurrent SCCs.
- Hollander–Solomyak [16] extend BD to all letter pairs in the two-letter case, settling Conjecture 6.13 for $|\mathcal{A}| = 2$. This is the only alphabet size in which the conjecture is fully established.
- Barge [6], Akiyama–Lee [2], and others establish Conjecture 6.13 for β-substitutions and certain self-similar families.
- For $|\mathcal{A}| = 3$, Proposition 6.4 gives $\mathrm{PF}(N_{\mathcal{C}}) \geq \beta|\beta_2|$ for any *closed* recurrent SCC with nonzero dominant wedge; existence of such an SCC (Observation 6.5) is conditional. This rules out degenerate spectral collapse on wedge-nonzero components but not the closed-nonproductive obstruction (cf. Remark 6.6).

- Empirically, the balanced-pair automaton was constructed for the complete census of irreducible Pisot specimens with $|\mathcal{A}| = 3$, $|\sigma(a)| \leq 3$ whose automaton is finite under symbolic inflate-and-cancel, together with randomized samples on $|\mathcal{A}| = 4$. No closed nonproductive SCC was found in any case; recurrent noncoincident SCCs that fail to be productive do occur but are invariably open (they escape to a productive component), so the balanced-pair algorithm still reaches coincidence. Independently, the strong-coincidence condition (Remark 7.4) holds on all 6,009 specimens tested across $|\mathcal{A}| = 3, 4, 5$. No counterexample to Conjecture 6.13 is known.

- The witness depth (smallest n realizing strong coincidence on the worst letter pair) is small in random samples but is *not* uniformly bounded, in two independent ways. First, it is unbounded in alphabet size: the substitution σ_n on $\{0, \ldots, n-1\}$ with $\sigma_n(i) = (i{+}1)\,0$ for $i < n-1$ and $\sigma_n(n{-}1) = 0$ is primitive Pisot with irreducible characteristic polynomial (the n-bonacci polynomial, $\beta \to 2$), and its worst-pair witness depth is exactly $n{+}1$; this is a clean explicit family, though not the extremal one. Second, and dominantly, depth is driven by proximity of β to 1: the deepest specimens observed at each alphabet size occur at the smallest Pisot β reachable there (e.g. depth 15 at $\beta \approx 1.3247$ for $|\mathcal{A}| = 3$, depth 18 at $\beta \approx 1.4433$ for $|\mathcal{A}| = 5$), and the two effects compound. Small Pisot β near 1 are sparse by the structure of the Pisot set, so such specimens are rare but real. In random samples with $\beta \geq 1.5$, mean depth is about 3–4 and $\det M_\sigma$ is not a driver (unimodular and non-unimodular subsamples are indistinguishable). No witness-depth bound uniform in $|\mathcal{A}|$ can hold; a uniform bound would have been strictly stronger than Conjecture 6.13, and is ruled out by the family above.

Remark 6.15 (Routes that do not close the conjecture). Three natural attack routes are insufficient:

(i) *Cycle displacement nonvanishing.* The flipped-Tribonacci computation (§6.1) refutes the statement "noncoincident $\mathcal{B}_\sigma$-cycle $\Rightarrow \delta(\gamma) \neq 0$."

(ii) *Mossé desubstitution descent.* A recurrent SCC is closed under both σ-inflation and Mossé desubstitution; descent stays within the SCC and produces no smaller balanced-pair state.

(iii) *Pure algebraic constraints on $N_\mathcal{C}$.* Lemma 6.2 gives $|\mathcal{C}| \geq |\mathcal{A}|$ and $\operatorname{spec}(M_\sigma) \subseteq \operatorname{spec}(N_\mathcal{C})$, but the pair $N_\mathcal{C} = M_\sigma$ with $P = \mathrm{Id}$ satisfies the algebraic constraints of (2) and is consistent with $\mathrm{PF}(N_\mathcal{C}) = \beta$. The combinatorial diagonal-free condition (Definition 6.12) must be used; the algebra alone is insufficient. Proposition 6.4 sharpens the spectral picture (alphabet 3), but $\beta|\beta_2| < \beta$, so Proposition 6.4 alone cannot close the conjecture (Remark 6.6).

A self-contained proof of Conjecture 6.13 therefore requires combinatorial input beyond the linear-algebraic and spectral-lower-bound structure isolated above. The natural candidate is a Pisot-shadowing pigeonhole argument lifting BD's letter-pair construction to arbitrary recurrent SCCs.

Remark 6.16 (Productivity is an SCC-level property). The diagonal-free condition of Definition 6.12 cannot be reduced to a per-state condition. A computational survey of irreducible Pisot specimens with $|\mathcal{A}| = 3$ finds that individual states s for which no interior zero-return of Δ_s is followed by a diagonal letter pair are common: a single balanced pair can fail to produce a coincidence sibling from any interior balanced sub-factorization of $\sigma(s)$. In every recurrent noncoincident SCC examined the SCC as a whole was nonetheless productive, with the producing coincidence supplied by some state of the SCC and propagated by strong connectedness. This is consistent with Conjecture 6.13 being genuinely a statement about SCCs rather than individual states, and indicates that any proof must use strong connectedness (or the Pisot dynamics across the whole orbit) and cannot proceed state-by-state.

Theorem 6.17 (Conditional main theorem)**.** *Conditional on G1 (Hypothesis 4.10, finiteness of $\mathcal{B}_\sigma$) and on Conjecture 6.13, every primitive Pisot substitution with irreducible characteristic polynomial gives rise to a tiling dynamical system with pure discrete spectrum.*

Proof. $\mathcal{B}_\sigma$ is finite (Hypothesis 4.10, G1). By Proposition 6.14, conditional on Conjecture 6.13, for every pair $a \neq b$ the BPA expansion of (ab, ba) contains a coincidence sibling at some iterate, i.e., the balanced-pair algorithm for (ab, ba) terminates with coincidence in the sense of Theorem 1.1. The bridge then delivers pure discrete spectrum. □

Part III: Assembly

7 The Pisot substitution conjecture

Theorem 7.1. *Let σ be a primitive Pisot substitution on a finite alphabet whose incidence matrix has irreducible characteristic polynomial, and suppose $\mathcal{B}_\sigma$ is finite* (Hypothesis 4.10, G1)*. Then, conditional on the SCC Producer Theorem (Conjecture 6.13), the associated tiling dynamical system has pure discrete spectrum.*

Proof. This is Theorem 6.17. □

Remark 7.2 (Inputs)*.* The proof uses four distinct inputs:

(i) The defect theorem [13], combined with $\det M_\sigma \neq 0$, for unique decodability (Theorem 3.3) and its power-iterated version (Corollary 3.4).

(ii) Mossé recognizability [19, 20], from primitivity and aperiodicity, for the supertile hierarchy on Ω_σ (Theorem 2.3) and hence for local witness injectivity (Theorem 5.3).

(iii) Pisot growth $\beta > 1$, for the spectral characterization of mass leakage (Lemma 6.1).

(iv) The SCC Producer Theorem (Conjecture 6.13). This is settled for $|\mathcal{A}| = 2$ by Hollander–Solomyak [16], partially by Barge–Diamond [10] (one seed), and unconditionally for β-substitutions and certain self-similar families by [6, 2]; the general case is open. The mass-balance and algebraic-structure lemmas (§§6.2–6.3) reformulate the open content but do not close it. For $|\mathcal{A}| = 3$, Proposition 6.4 contributes a spectral lower bound on the within-SCC matrix that is informative but, by Remark 6.6, insufficient to close the conjecture even on alphabet 3.

The bridge from balanced-pair termination to pure discrete spectrum is Barge–Štimac–Williams [12] (which closes the gap in [11] flagged in Remark 1.2). Neither full recognizability [15] nor any residual-graph analysis is required. The defect theorem gives UD in five lines of elementary combinatorics on words and linear algebra.

Remark 7.3 (Architectural contribution)*.* The substantive mathematical simplifications proved in this paper are:

- UD via the defect theorem (Theorem 3.3): five lines from $\det M_\sigma \neq 0$, using no substitution-specific machinery.
- Explicit padding bound via the phase automaton (Observation 4.7): $D(\sigma) \leq |\mathcal{A}|^2 |\sigma|^2_{\max}$, replacing the unspecified "padding bound" constant in earlier formulations.
- Local witness injectivity (Theorem 5.3): replaces the multi-page recognizability-based key lemma of Barge–Diamond [10] and Barge–Kwapisz [11], in its correct local form.

- Mass-balance reformulation of the Level-3 closure (Lemmas 6.1, 6.2): the open content is now sharply identified as the SCC Producer Theorem (Conjecture 6.13), with the cycle-exclusion target of earlier drafts shown to be false (§6.1) and the closed-nonproductive case forced into a constrained algebraic structure ($N_{\mathcal{C}}$ contains M_σ as an invariant sub-block, $|\mathcal{C}| \geq |\mathcal{A}|$) that consumes part of the difficulty.

- Alphabet-3 spectral mechanism (Proposition 6.4): any *closed* recurrent SCC with nonzero dominant wedge has $\mathrm{PF}(N_{\mathcal{C}}) \geq \beta|\beta_2|$; existence of a qualifying SCC is conditional (Observation 6.5).

- Cohomological consolidation (§6.5): the closed-nonproductive SCC is identified with the transversal substitution $\Sigma_{\mathcal{C}}$ on the Barge–Diamond side, yielding the unconditional leg-factor coincidence-rank floor $\mathrm{cr}(\Sigma_{\mathcal{C}}) \geq 2$ (Corollary 6.8), the triple mass-balance ceiling (Lemma 6.9), and a second conditional route to unimodular pure discrete spectrum via homological-Pisotness of $\Sigma_{\mathcal{C}}$ together with the Coincidence Rank Conjecture (Proposition 6.10).

The remaining open content is the Pisot-shadowing combinatorial step needed to rule out diagonal-free zero-return systems.

Remark 7.4 (Computation). The strong-coincidence condition that is the productivity content of the SCC Producer Theorem (Conjecture 6.13)—that for each letter pair $a \neq b$ some iterate $\sigma^n(a), \sigma^n(b)$ shares a position with equal prefix Parikh vectors—was verified directly on 6,009 irreducible Pisot specimens: the complete census of all such substitutions with $|\mathcal{A}| = 3$, $|\sigma(a)| \leq 3$ (4,554 specimens), together with randomized samples on $|\mathcal{A}| = 4, 5$ (1,077 and 378 specimens), with zero failures. The test is performed on letter pairs directly and does not construct the balanced-pair automaton, so it includes 2,244 non-unimodular specimens ($|\det M_\sigma| \geq 2$) without difficulty. The maximal witness depth observed was 15, attained only at β equal to the plastic number (the smallest Pisot number); 99.7% of $|\mathcal{A}| = 3$ specimens coincide by depth 8. This checks the productivity content of Conjecture 6.13, not pure discrete spectrum itself, which (outside $|\mathcal{A}| = 2$) requires the bridge of Theorem 1.1; the two are not conflated (cf. §1.1).

Theorem 3.3 was verified on the complete census of 59,319 three-letter substitutions with $|\sigma(a)| \leq 3$: of these, 7,491 are non-UD (including duplicate-image specimens), and every non-UD specimen has $\det M_\sigma = 0$, consistent with the theorem. Across the 4,554 irreducible Pisot specimens in that census, the Sardinas–Patterson unique-decodability test and the criterion $\det M_\sigma \neq 0$ agreed in every case, with no mismatch.

The integral residual graph of Sardinas–Patterson provides an independent computational test: all Pisot specimens pass (empty forward-backward intersection at ε_0, so ε_0 cannot return to itself). Whether the specific structural fact that ε_0 lies outside every cycle of R_σ^{int} admits a direct combinatorial proof from $\det \neq 0$ alone is an interesting question in codes-and-automata, independent of the Pisot conjecture and not on its critical path.

For Proposition 6.4, an empirical sweep of 500 random PIP specimens on alphabet 3 found that all three length-2 swap pairs had nonzero dominant projection in every case (500/500, minimum projection norm 0.690). This is consistent with Lemma 6.3, which guarantees only that at least one of the three has nonzero projection.

Remark 7.5 (Companion note). The proof of Proposition 6.4 (sketched in §6.4) is given in detail in a companion note documenting the v33–v34 development: Inter-Block Cancellation Lemma (whole-pair K_2-conservation under σ-decomposition), Dominant K_2-Source Lemma (Lemma 6.3), and the path-counting plus SCC-condensation argument. The finite-closure input used inside the proof is supplied by Hypothesis 4.10 of the present paper.

8 Example

Example 8.1. $\sigma : 1 \mapsto 2,\ 2 \mapsto 3,\ 3 \mapsto 12$. $\beta \approx 1.3247$, $\det M_\sigma = 1$, characteristic polynomial $x^3 - x - 1$ (irreducible over $\mathbb{Q}$). Images $\{2, 3, 12\}$ form a prefix code, hence UD (defect-theorem argument collapses: $|H| = |X| = 3$). The balanced-pair automaton has 44 states; coincidence reached within 12 substitution steps. The three length-2 swap pairs s_{12}, s_{13}, s_{23} all have nonzero dominant $\Lambda^2 M_\sigma$-projection (computed numerically), consistent with Lemma 6.3.

References

[1] S. Akiyama, M. Barge, V. Berthé, J.-Y. Lee, A. Siegel, *On the Pisot substitution conjecture*, in *Mathematics of Aperiodic Order*, Progr. Math. **309**, Birkhäuser, 2015, 33–72.

[2] S. Akiyama, J.-Y. Lee, *Algorithm for determining pure pointedness of self-affine tilings*, Adv. Math. **226** (2011), 2855–2883.

[3] S. Akiyama, J.-Y. Lee, *Overlap coincidence to strong coincidence in substitution tiling dynamics*, European J. Combin. **39** (2014), 233–243.

[4] S. Balchin, D. Rust, *Computations for symbolic substitutions*, J. Integer Seq. **20** (2017), Article 17.4.1.

[5] M. Barge, *Factors of Pisot tiling spaces and the coincidence rank conjecture*, Bull. Soc. Math. France **143** (2015), 357–381.

[6] M. Barge, *The Pisot conjecture for β-substitutions*, Ergodic Theory Dynam. Systems **38** (2018), 444–472.

[7] M. Barge, H. Bruin, L. Jones, L. Sadun, *Homological Pisot substitutions and exact regularity*, Israel J. Math. **188** (2012), 281–300.

[8] M. Barge, B. Diamond, *Cohomology in one-dimensional substitution tiling spaces*, Proc. Amer. Math. Soc. **136** (2008), 2183–2191.

[9] L. Sadun, *Cohomology of hierarchical tilings*, in *Mathematics of Aperiodic Order*, Progr. Math. **309**, Birkhäuser (2015), 73–104.

[10] M. Barge, B. Diamond, *Coincidence for substitutions of Pisot type*, Bull. Soc. Math. France **130** (2002), 619–626.

[11] M. Barge, J. Kwapisz, *Geometric theory of unimodular Pisot substitutions*, Amer. J. Math. **128** (2006), 1219–1282.

[12] M. Barge, S. Štimac, R. F. Williams, *Pure discrete spectrum in substitution tiling spaces*, Discrete Contin. Dyn. Syst. **33** (2013), 579–597.

[13] J. Berstel, D. Perrin, J.-F. Perrot, A. Restivo, *Sur le théorème du défaut*, J. Algebra **60** (1979), 169–180.

[14] J. Berstel, D. Perrin, C. Reutenauer, *Codes and Automata*, Encyclopedia of Mathematics and its Applications **129**, Cambridge University Press, 2009.

[15] V. Berthé, W. Steiner, J. M. Thuswaldner, R. Yassawi, *Recognizability for sequences of morphisms*, Ergodic Theory Dynam. Systems **39** (2019), 2896–2931.

[16] M. Hollander, B. Solomyak, *Two-symbol Pisot substitutions have pure discrete spectrum*, Ergodic Theory Dynam. Systems **23** (2003), 533–540.

[17] S. Ito, H. Rao, *Atomic surfaces, tilings and coincidence I: irreducible case*, Israel J. Math. **153** (2006), 129–155.

[18] P. Mercat, S. Akiyama, *Yet another characterization of the Pisot substitution conjecture*, in *Substitution and Tiling Dynamics*, Lecture Notes in Math. **2273**, Springer, 2020, 397–448.

[19] B. Mossé, *Puissances de mots et reconnaissabilité des points fixes d'une substitution*, Theoret. Comput. Sci. **99** (1992), 327–334.

[20] B. Mossé, *Reconnaissabilité des substitutions et complexité des suites automatiques*, Bull. Soc. Math. France **124** (1996), 329–346.

[21] K. Nakaishi, *Pisot conjecture and Rauzy fractals*, arXiv:1610.03146 (2016–2023, withdrawn).

[22] K. Nakaishi, *Pisot substitution conjecture and Rauzy fractals*, arXiv:2401.07771 (2024).

[23] M. Queffélec, *Substitution Dynamical Systems—Spectral Analysis*, 2nd ed., Lecture Notes in Math. **1294**, Springer, 2010.